One-head attention pooling adds little beyond a deeper MLP in a Qwen2.5-VL region probe

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https://github.com/nkn002/visual-readout

We test whether attention pooling over the tokens under a box improves the HW2 probe, which names a Visual Genome box's content from Qwen2.5-VL-7B's hidden state 21. Our design (one-head attention and a two-hidden-layer MLP) beats HW2 by +0.71 points, but the same MLP on the mean pool already gains +0.48. Eight heads gain about one point, yet part of that gain appears on one-token boxes, where attention cannot act, and person recall does not improve.

1 Hypothesis, committed before training

Problem. The HW2 probe averages the L tokens under a box and feeds the 3,584-d mean to an MLP ($3584 \to 512 \to \text{ReLU} \to 20$) that predicts one of 20 categories. Its weakest category is person (50% recall), whose boxes often show mostly a shirt, a face, or an arm.

Design. We replace the mean with attention pooling and add a second hidden layer ($3584 \to 512 \to \text{ReLU} \to 512 \to \text{ReLU} \to 20$). The pooled vector is $\sum_i a_i x_i$ with $a_i = \text{softmax}_i\left(q^\top W_k \hat{x}_i / \sqrt{64}\right)$, where x_i are the tokens under the box and $\hat{x}_i$ their standardized values. The query q starts at zero, so training starts from the exact mean pool. An 8-head variant gives each head its own query and its own 448-d slice of the vector.

Predictions. Learned weights can favor object tokens over background tokens. Before any arm trained (commit `c03e240` at 15:02; first result at 15:37), we predicted: (P1) no change on one-token boxes; (P2) gains for boxes of 10 or more tokens, growing with size; (P3) the largest recall gains for person, limb, and clothing; (P4) a second hidden layer on the mean pool adds overfitting, not accuracy; and (P5) 8 heads match or beat 1 head. We also named four failure modes, F1 to F4 (Tab. 2).

2 Setup

We cross pooling (mean, attention with 1 head, attention with 8 heads) with depth (1 or 2 hidden layers) on the HW2 split of 50,000 regions (2,500 per category; 39,968 train, 5,075 validation, and 4,957 test, split by image). Every arm searches the same grid (Adam; learning rate {3e-4, 1e-3} $\times$ weight decay {1e-4, 1e-3}; batch 256; dropout 0.5; early stopping), keeps its best setting on validation accuracy, and reruns it with 3 seeds. We score the test set once per arm and report paired-bootstrap 95% CIs over test regions, which leave out training-seed variance. The 843 one-token test boxes act as a placebo: there attention returns the token itself, so the pooled input equals HW2's.

3 Results

The second hidden layer, not one-head attention, carries most of the proposed design's gain. The proposed design beats HW2 by +0.71 points (Tab. 1). Yet the same MLP on the mean pool already gains +0.48, so one-head attention adds only +0.24 at the same depth, with a CI that includes zero. In contrast, eight heads beat the mean pool at both depths (+0.86 and +0.97), and the best arm reaches 75.07%.

One head gains where predicted, but eight heads also gain where attention cannot act. At HW2's depth, one head gains +0.08 points on one-token boxes and +1.80 on boxes of 50

Table 1: The proposed design (gray) beats HW2 by +0.71 points, but depth alone gives +0.48. Test accuracy (%) is the mean ± std over 3 seeds; Δ is the gain in points with its 95% CI.

Pooling	Hidden	Test acc.	Δ vs. HW2	Δ vs. mean, same depth	Epoch (s)
Mean (HW2)	1	73.62 ± 0.38	–	–	0.09
Mean	2	74.10 ± 0.21	+0.48 [+0.01, +0.95]	–	0.21
Attention, 1 head	1	74.10 ± 0.15	+0.48 [−0.09, +1.06]	+0.48 [−0.09, +1.06]	0.77
Attention, 1 head	2	74.33 ± 0.08	+0.71 [+0.05, +1.32]	+0.24 [−0.40, +0.80]	0.69
Attention, 8 heads	1	74.48 ± 0.17	+0.86 [+0.26, +1.43]	+0.86 [+0.26, +1.43]	1.09
Attention, 8 heads	2	**75.07** ± 0.25	+1.45 [+0.82, +2.06]	+0.97 [+0.40, +1.52]	1.23

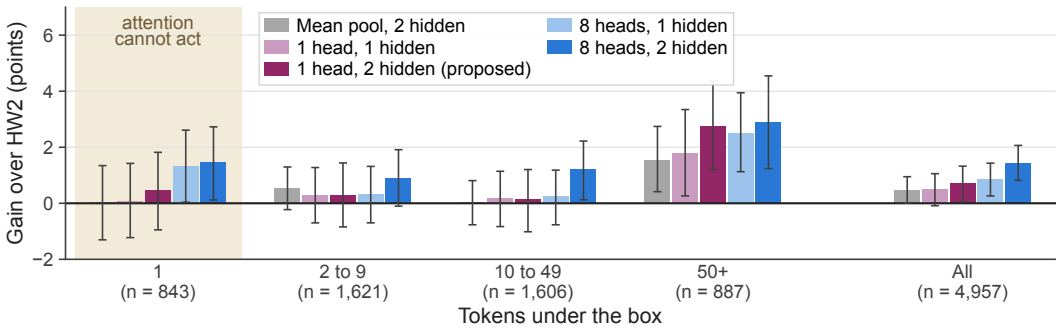


Figure 1: Part of the 8-head gain appears on one-token boxes, where attention cannot act (shaded). Bars give the test-accuracy gain over HW2 per bin of tokens under the box (3-seed mean); whiskers give 95% CIs.

or more tokens (Fig. 1), as P1 and P2 predicted. However, depth alone also gains +1.54 on these large boxes. Interestingly, eight heads at HW2's depth gain +1.30 [+0.04, +2.61] on one-token boxes, where their input equals HW2's, more than their +0.77 on larger boxes. This borderline result (one of about 30 box-size contrasts) suggests that part of the eight-head gain comes from the jointly trained MLP or from seed variance (HW2's seeds span 73.09% to 73.90%), not from re-weighting tokens.

Attention does not fix person recall. Person recall changes by −2.3 points [−5.9, +1.2] for the proposed design and by −0.3 for the best arm, while clothing, window and door, street and road, and furniture gain. The weights stay diffuse (normalized entropy 0.90 to 0.95, where 1 is uniform), and attention costs 3 to 12× more time per epoch.

Conclusion. Our design gains +0.71 points over HW2, mostly through its second hidden layer, and does not fix person recall (Tab. 2). Re-scoring each attention probe with uniform weights (not run) would separate the pooling from the jointly trained MLP, and more seeds would put seed variance into the CIs.

Table 2: The data support P5, partly support P2 and P4, contradict P1 for eight heads, and do not support P3 for person. Brackets give 95% CIs in points.

Prediction or failure mode	Observation	Verdict
P1. 1-token boxes unchanged	1 head +0.08; 8 heads +1.30 [+0.04, +2.61]	Only for 1 head
P2. Gain grows with box size	50+ tokens: attention +1.8 to +2.9; depth +1.5	Partly; depth too
P3. Person, limb, clothing	Person −2.3; clothing +3.2 and +6.8	Not for person
P4. Depth: overfitting only	Train 86.4% vs. 82.4%; test +0.48 [+0.01, +0.95]	Partly refuted
P5. 8 heads ≥ 1 head	+0.38 (1 hidden); +0.73 [+0.24, +1.23] (2 hidden)	Supported
F1. Weights stay uniform	Entropy 0.90 to 0.95 (1 is uniform)	No; diffuse
F2. Position lock	First token on top: 15 to 18% vs. 13 to 14%	Weak at most
F3. Overfitting	Train-val. gap 9.4 (HW2), 13.3 (8 heads, 1 hidden)	In one arm
F4. Decay, outliers, scaling	Not measured	Not tested